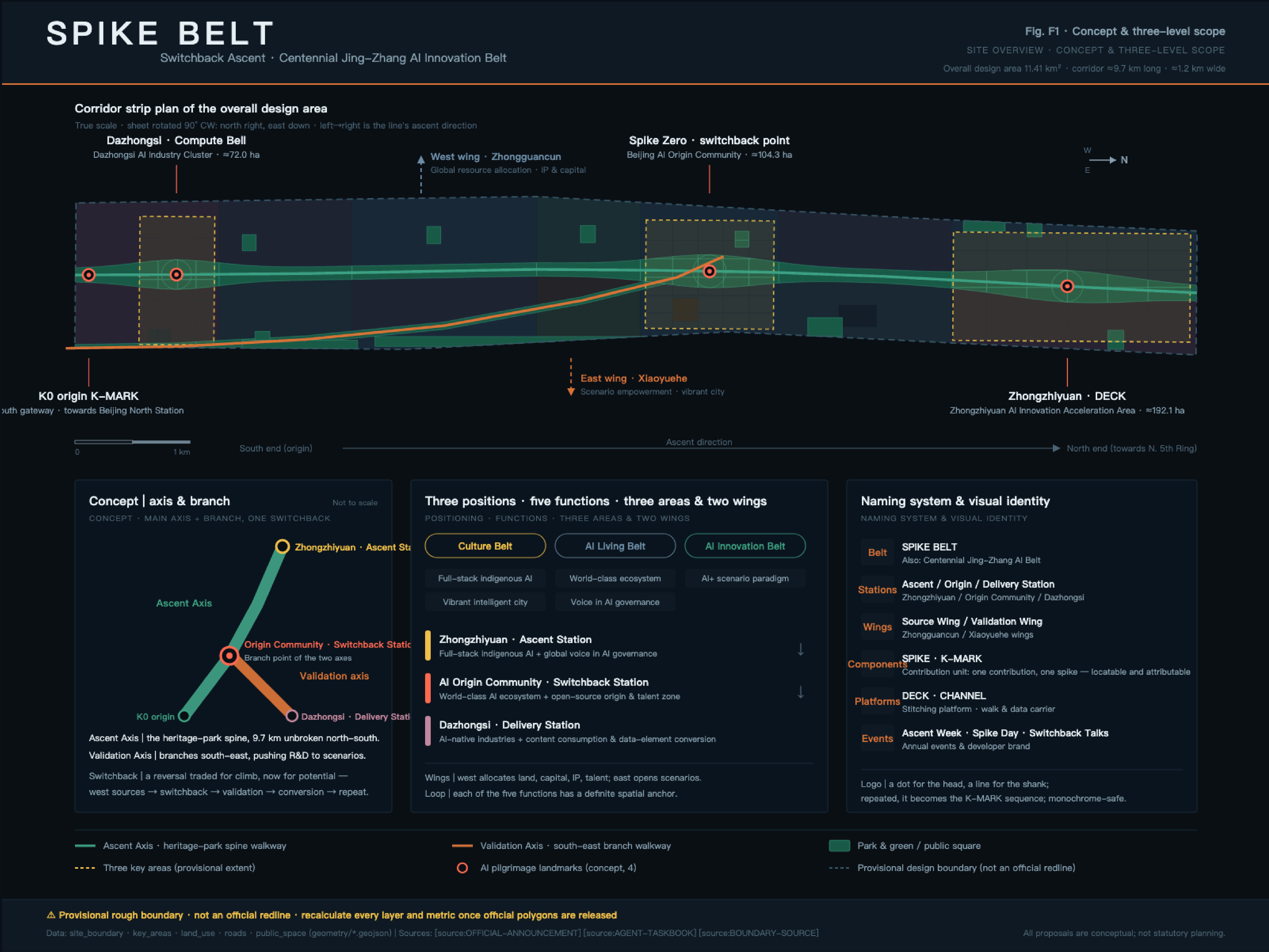




Spatial structure & land use

From one axis, one branch, three switchbacks, two wings — to 187 parcels (90 blocks in the key areas)

Fig. F2 · Structure & land use

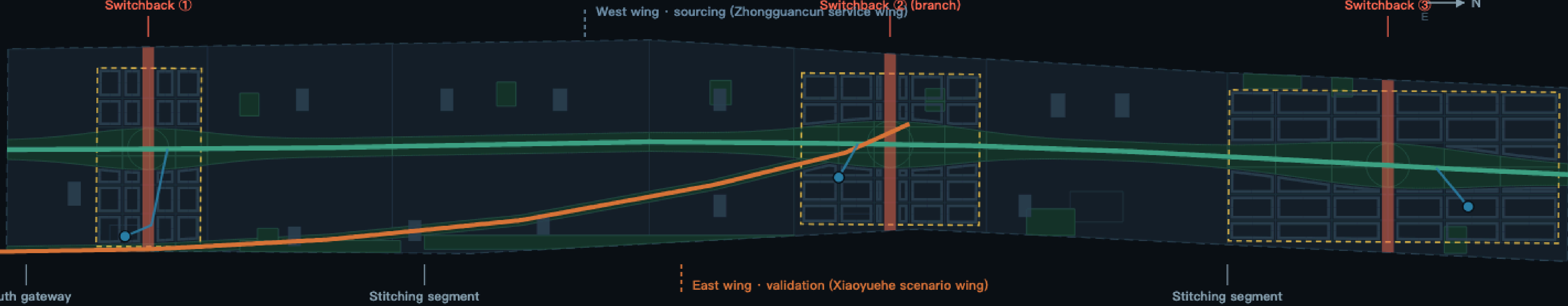
SPATIAL STRUCTURE → LAND USE

Land use fully covers the submitted boundary without overlap · recalculated in EPSG:4548

A | Structure

STRUCTURE — spine, switchbacks, stitching, gateways, wings

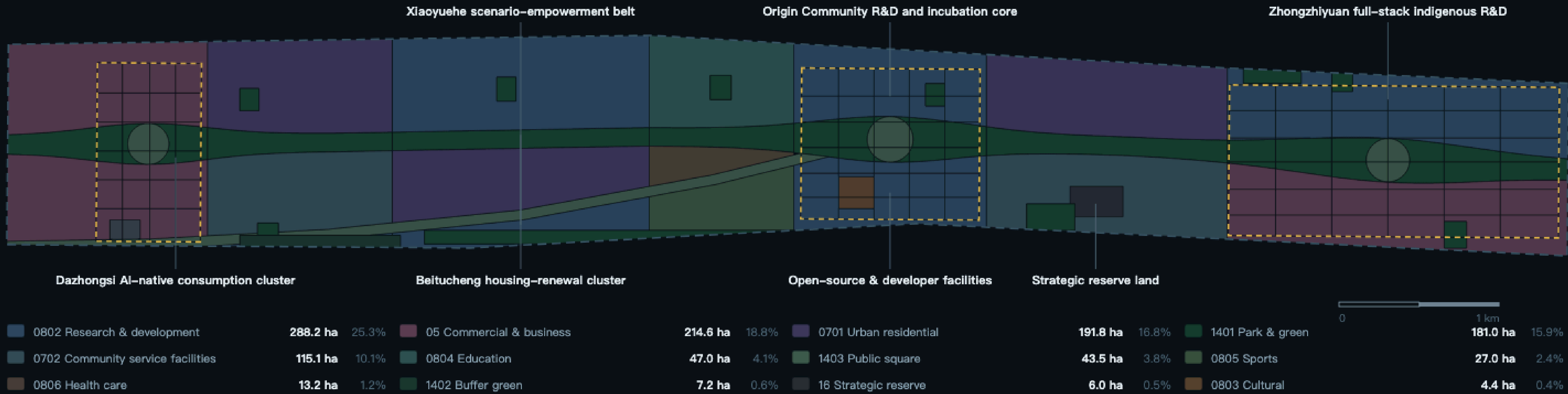
Structure is not a new redline; it turns the three-level scope into locatable axes, nodes and areas.



B | Land use

LAND USE — coded per the national land-use classification guide

Land use is a concept proposal, not a statutory classification; intensity and height await official regulatory conditions.



Code	Category	ha	Share
0802	Research & development	288.2	25.25%
05	Commercial & business	214.6	18.80%
0701	Urban residential	191.8	16.81%
1401	Park & green	181.0	15.86%
0702	Community service facilities	115.1	10.09%
0804	Education	47.0	4.12%
1403	Public square	43.5	3.81%
0805	Sports	27.0	2.36%
0806	Health care	13.2	1.16%
Total 187		1 141.3	100%

Retain / renovate / rebuild

Retain 65 · Renovate 94 · Rebuild 32 · New build 129 — 320 footprints, 94.29 ha.

On a corridor of existing stock, the default move should be renovation, not demolition.

Intensity and height are withheld for lack of approved regulatory conditions and stay unknown in metrics.json; the retain/renovate/rebuild split needs ownership checks and statutory process.

Three-level scope transmission

THREE-LEVEL SCOPE TRANSMISSION

Coordinated research area 43.6 km²

Industrial ecosystem + future urban form

Overall design area 11.41 km²

Structure / land use / mobility / blue-green / character

Key detailed-design areas 368.4 ha

Three detailed designs & delivery handles

Upper decides, lower draws, upper re-checks.

Land-use composition (recalculated)

LAND USE COMPOSITION (EPSG:4548)

Land Use Code	Land Use Name	Area (ha)	Percentage (%)
0802	Research & development	288.2	25.25%
05	Commercial & business	214.6	18.80%
0701	Urban residential	191.8	16.81%
1401	Park & green	181.0	15.86%
0702	Community service facilities	115.1	10.09%
0804	Education	47.0	4.12%
1403	Public square	43.5	3.81%
0805	Sports	27.0	2.36%

Key structural moves

KEY STRUCTURAL MOVES

- N-S link** The spine runs 9.7 km; the CHANNEL walkway never breaks.
- E-W stitch** Three DECKs span the corridor, rejoining both sides.
- Sides** West sources R&D; east validates scenarios and pilots.
- Wedge** Two axes enclose a southern wedge district.
- Reserve** 6.0 ha reserved in the north keeps options open.

△ Provisional rough boundary · not an official redline · recalculate every layer and metric once official polygons are released

Data: geometry/land_use.geojson (187 parcels) · phasing.geojson · roads.geojson · key_areas.geojson | Standards: [standard:MNR-LAND-USE-CLASSIFICATION-GUIDE] [standard:MOHURD-CONTROL-DETAILED-PLANNING]

Fig. F2 | A structure; B land use (187 parcels, full cover, no overlap).

Mobility · blue-green public space · AI scenario nodes

CHANNEL carries north-south, DECKs stitch east-west, components anchor scenarios

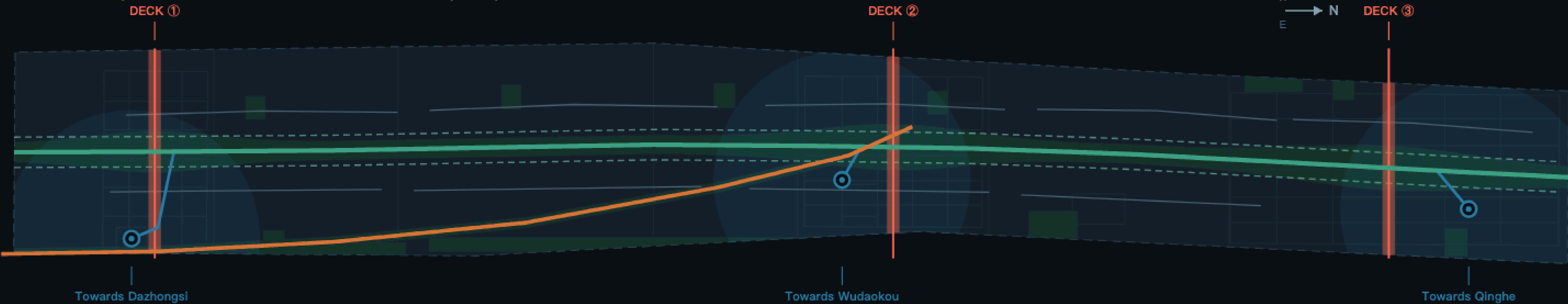
Fig. F4 · Mobility & blue-green

MOBILITY · BLUE-GREEN · AI SCENARIO NODES

Green 16.49% · public space 7.23% · spine walkway 9 728.5 m

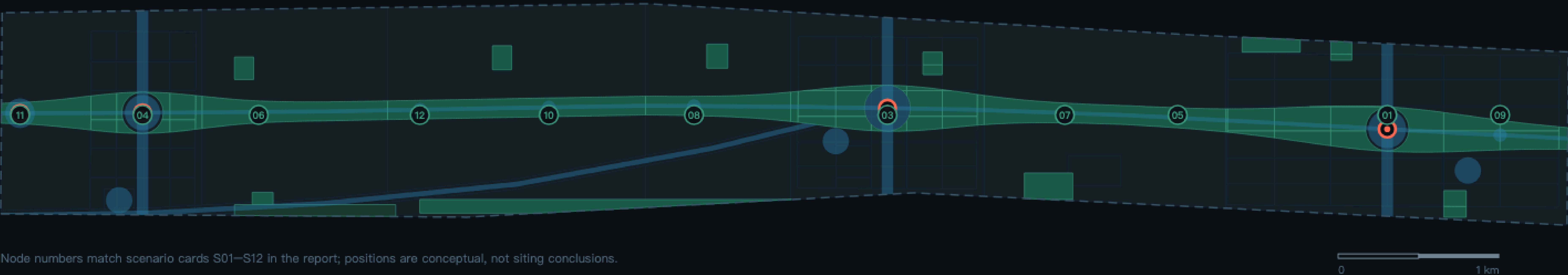
A | Mobility, transit interchange & walking

MOBILITY — all alignments and nodes are schematic; not road redlines or engineering conclusions



B | Blue-green, public space & AI scenario nodes

BLUE-GREEN & PUBLIC SPACE — the two overlap inside the spine; each ratio is recalculated independently



Node numbers match scenario cards S01–S12 in the report; positions are conceptual, not siting conclusions.

DECK section | east-west stitching + north-south continuity

SWITCHBACK DECK — SCHEMATIC SECTION, NOT TO SCALE



Transit interchange & slow mobility

TRANSIT & SLOW MOBILITY

- 3 Conceptual nodes and squares (not stations)
- 800 m Catchment rings used to check facilities and walking
- 3 DECK east-west walkways (stitching the break)
- 9 Suggested local-street loops (5 west / 4 east)
- 2 Cycle tracks paralleling the spine

Based density and parking await official data

Blue-green & public space recalculated

RECALCULATED FROM GEOMETRY

Green ratio green_ratio	16.49 %
58 green features / site area	
Public-space ratio	7.23 %
19 public-space features, dissolved before计算	
Park & green	181.0 ha
1401 Park & green (spine + pocket parks)	
Public square	43.5 ha
1403 Three DECK squares + validation-axis corridor	
AI pilgrimage landmarks	4
Spike Zero / DECK / Compute Bell / K0 marker	

The two ratios overlap in space and must not be added

Mobility · blue–green public space · AI scenario nodes

CHANNEL carries north–south, DECKs stitch east–west, components anchor scenarios

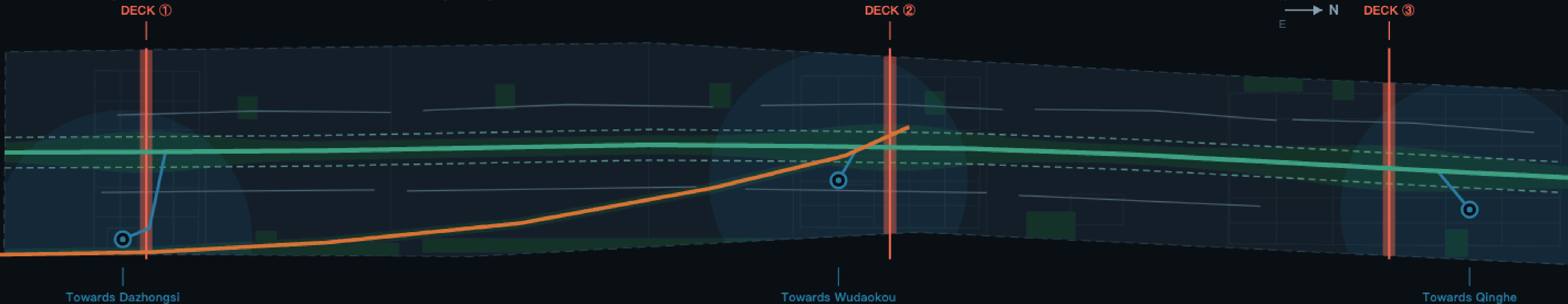
Fig. F4 · Mobility & blue–green

MOBILITY · BLUE–GREEN · AI SCENARIO NODES

Green 16.49% · public space 7.23% · spine walkway 9 728.5 m

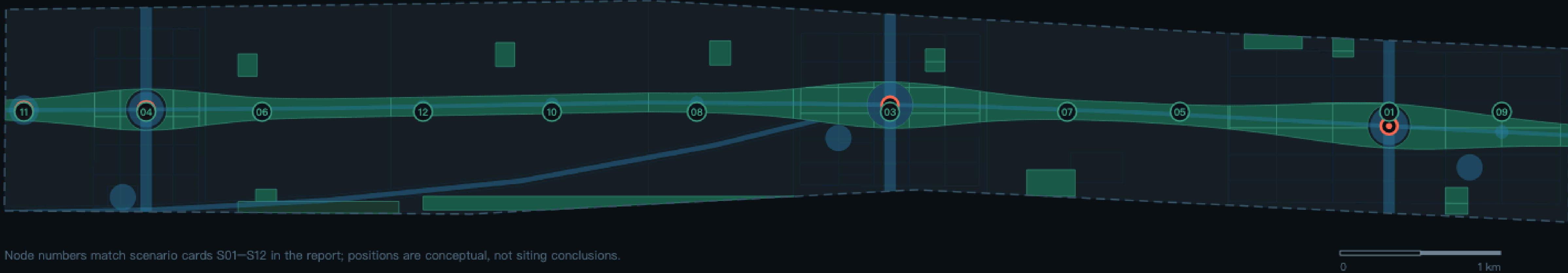
A | Mobility, transit interchange & walking

MOBILITY — all alignments and nodes are schematic; not road redlines or engineering conclusions



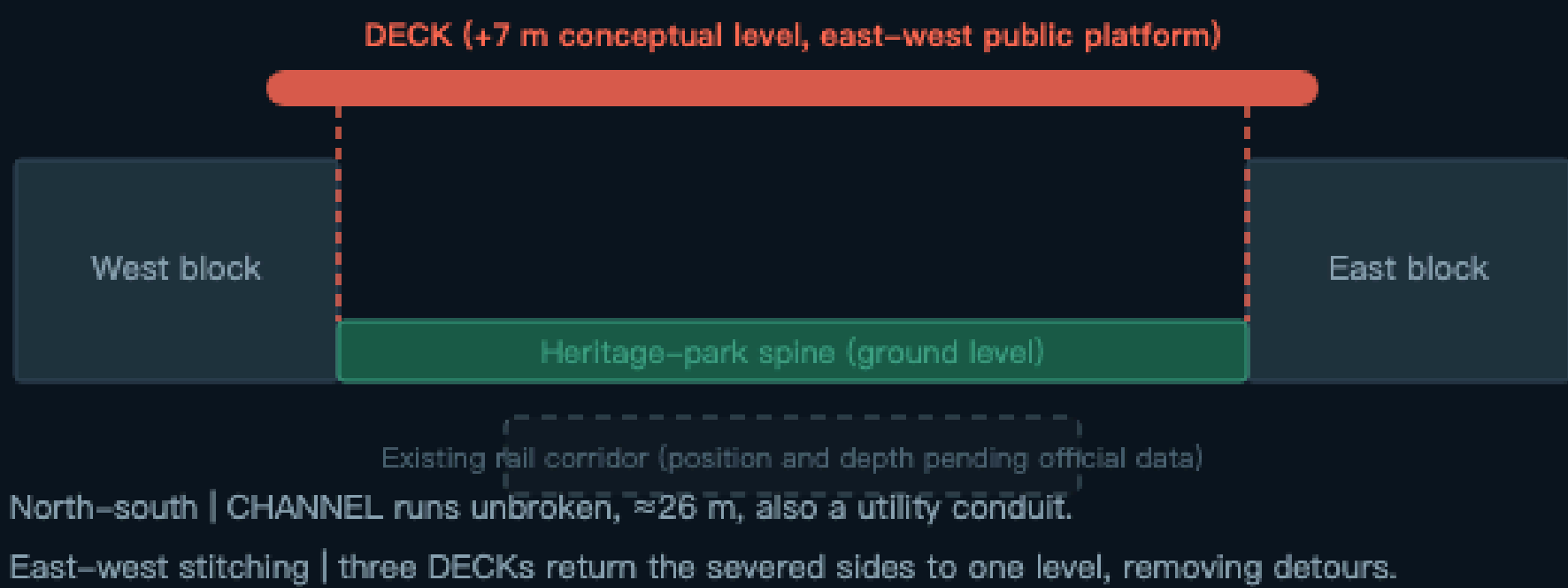
B | Blue–green, public space & AI scenario nodes

BLUE–GREEN & PUBLIC SPACE — the two overlap inside the spine; each ratio is recalculated independently



DECK section | east–west stitching + north–south continuity

SWITCHBACK DECK — SCHEMATIC SECTION, NOT TO SCALE



Transit interchange & slow mobility

TRANSIT & SLOW MOBILITY

- 3 Conceptual nodes and squares (not stations)
- 800 m Catchment rings used to check facilities and walking
- 3 DECK east–west walkways (stitching the break)
- 9 Suggested local–street loops (5 west / 4 east)
- 2 Cycle tracks paralleling the spine

Road density and parking await official data.

Blue–green & public space recalculated

RECALCULATED FROM GEOMETRY

Green ratio green_ratio	16.49 %
58 green features / site area	
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19 public-space features, dissolved before计算	
Park & green	181.0 ha
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Public square	43.5 ha
1403 Three DECK squares + validation-axis corridor	
AI pilgrimage landmarks	4
Spike Zero / DECK / Compute Bell / K0 marker	
The two ratios overlap in space and must not be added.	

- Ascent Axis walkway
- Validation axis
- DECK east–west walkway
- Interchange links & nodes
- Cycle tracks
- Suggested local–street loops
- Green & open space
- Public space

△ Alignments, interchange nodes and DECK levels are schematic. Data: geometry/roads.geojson · green_space.geojson · public_space.geojson · constraints.geojson

Public-space component family

3 DECKs · 1 CHANNEL · 7 K-MARK squares · 3 station squares · 1 validation-axis corridor · 4 landmarks.

The family matters because it is replicable, extendable and phaseable: a new node draws from it rather than inventing a new language.

Four AI pilgrimage landmarks (conceptual)

Spike Zero A ring plaza on the branch point. When a railway is finished the last spike goes in; an open-source city never drives its last one.

Zhongzhiyuan DECK The platform and the viewable test ground beneath it.

Compute Bell Echoing the temple's bell motif, translating live scenario status into sound and light.

K0 origin K-MARK The belt's start marker and international gateway.

Form, siting and acoustic impact need heritage, landscape and environmental sign-off. Nothing is over-commercialised, and no unlicensed trademark, font, image or likeness is used.

Fig. F4-B | 58 green features derive from land-use classes 1401 / 1402, so the ratio can be checked two ways.

Three key detailed-design areas

Ascent · Origin · Delivery — three roles, strung into one loop by the two axes

Fig. F3 · Key areas

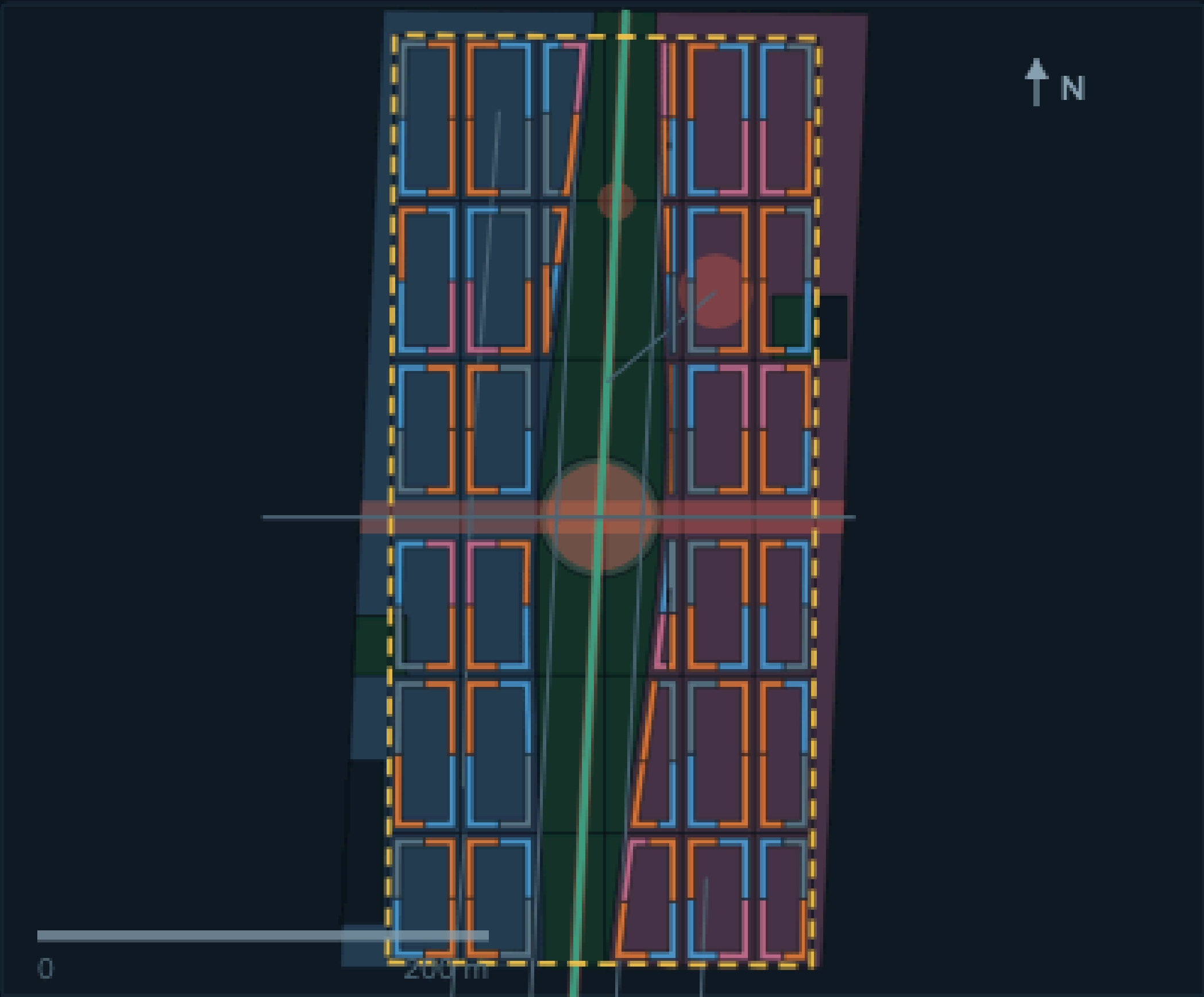
THREE KEY DETAILED-DESIGN AREAS

Three areas at one scale (1 : ≈5300) · north up · ≈368.4 ha total

① Zhongzhiyuan AI Innovation Acceleration Area

ASCENT STATION · ≈192.1 ha

Full-stack indigenous AI + global voice in AI governance



Spatial moves | SPATIAL MOVES

- **DECK ③ east-west stitching**
An east-west platform across the corridor, returning the Qinghe housing side and the
- **Outdoor test & validation ground**
Beneath the DECK sits a viewable test ground with seating — validation itself becomes
- **Buffer wedge + strategic reserve**
A buffer wedge along the 5th Ring; 6.0 ha reserved in the north keeps options for

Main renewal projects | PROJECTS

- Full-stack indigenous R&D complex (new)
- Open lab & standards testing building (new)
- Converted plant R&D building (renovate)
- Accelerator cluster (new)
- Talent housing (new)
- Exhibition & international exchange centre (new)

AI S01 test ground · S02 sandbox · S09 shuttle

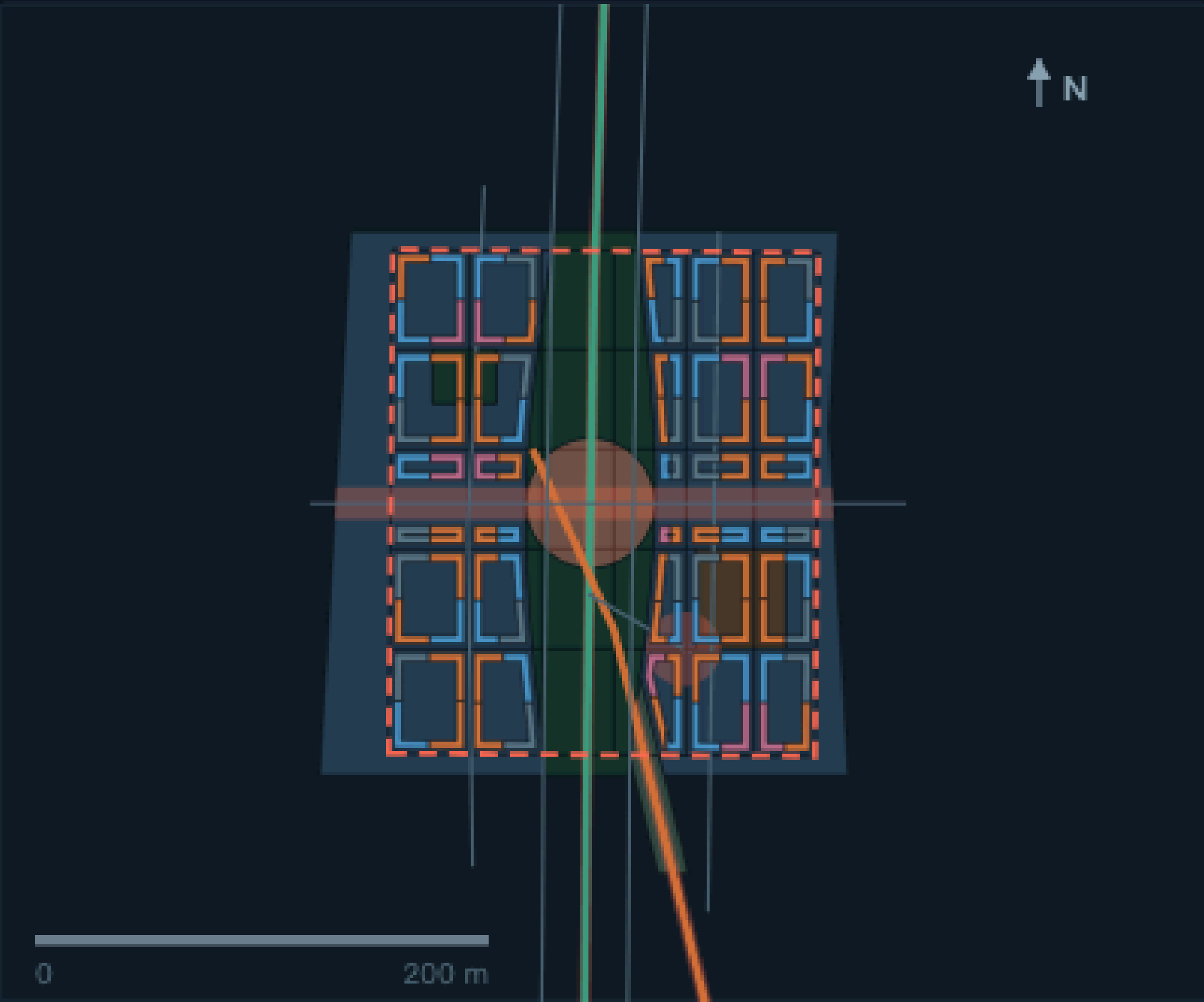
Pending conditions | PENDING

- Noise and buffer conditions along the 5th Ring pending
- Existing plant ownership and structural safety to verify
- Compute energy load and utility capacity need專業

② Beijing AI Origin Community

ORIGIN STATION · ≈104.3 ha

World-class AI ecosystem + open-source origin & talent zone



Spatial moves | SPATIAL MOVES

- **Spike Zero honour ring**
A ring plaza on the branch point; contributor spikes and GitHub handles engraved —
- **Campus-adjacent innovation interface**
Teaching, R&D and incubation share one block, turning "near campus" from a location claim
- **Tsinghuayuan Station cultural node**
A display and narrative start point on the railway remains; heritage authorities must

Main renewal projects | PROJECTS

- R&D and incubation building (new)
- Existing building to incubator (renovate)
- Open-source collaboration lab (new)
- Talent housing (new)
- Tsinghuayuan Station museum (heritage reuse · renovate)
- East renewal office cluster (rebuild)

AI S03 open-source · S05 guide · S07 services

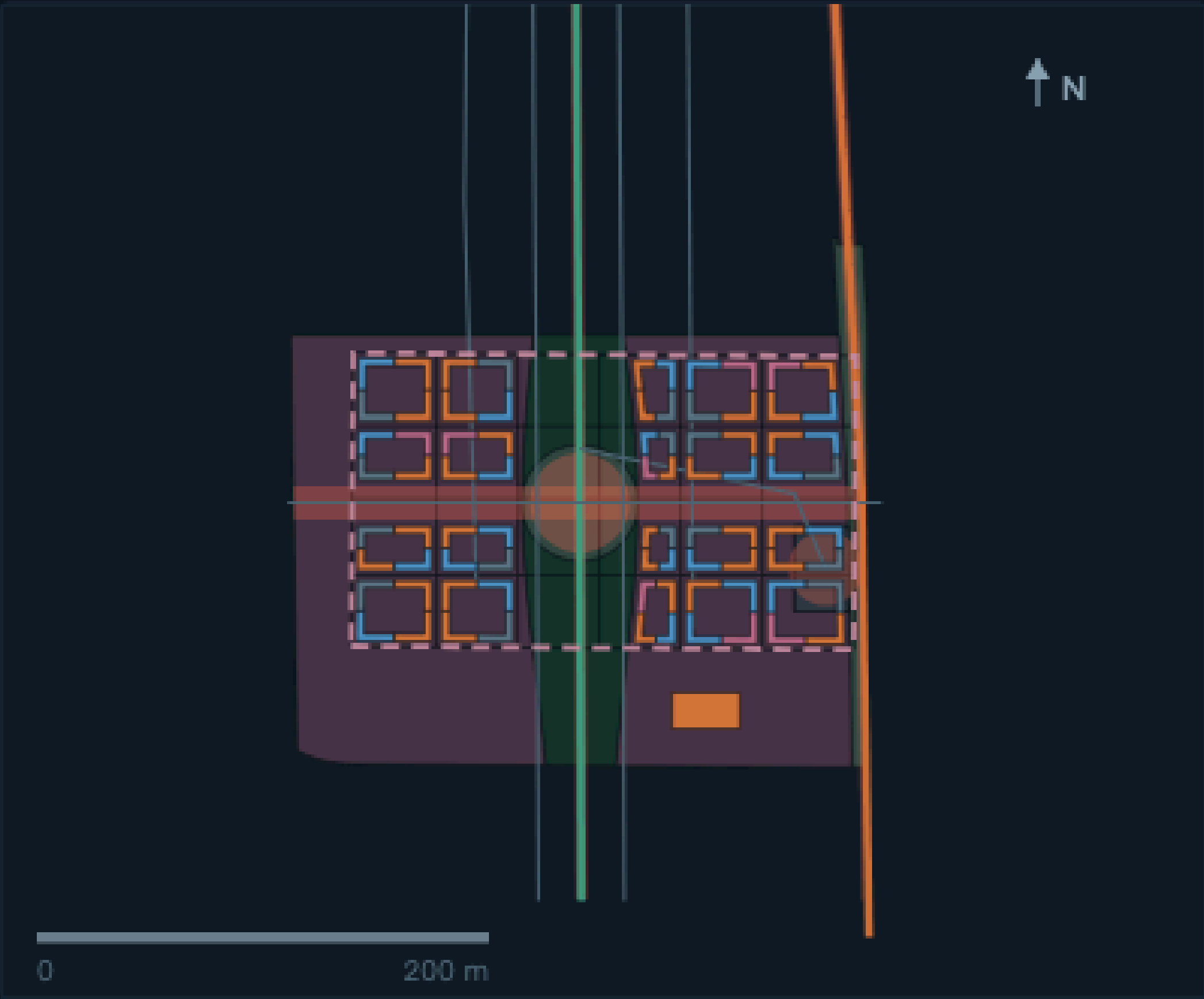
Pending conditions | PENDING

- Heritage protection and control zone pending
- University and institute land-ownership boundaries to verify
- Transit interchange towards Wudaokou pending

③ Dazhongsi AI Industry Cluster

DELIVERY STATION · ≈72.0 ha

AI-native industries + content consumption & data-element conversion



Spatial moves | SPATIAL MOVES

- **DECK ① four-quadrant stitching**
Stitching the four quadrants of the Dazhongsi junction and removing today's detours.
- **Compute Bell plaza**
A public installation plaza echoing the temple's bell motif, translating live
- **Integrated interchange site**
A 2.3 ha interchange site integrating rail, bus, walking and low-speed shuttle

Main renewal projects | PROJECTS

- AI-native consumption complex (new)
- Smart-device R&D building (new)
- West business renewal cluster (rebuild)
- West retail conversion (renovate)
- Integrated interchange hub (new)
- South gateway culture & visitor centre (new)

AI S04 robots · S06 devices · S12 data services

Pending conditions | PENDING

- Dazhongsi heritage protection extent pending
- Junction traffic organisation and signals need專業
- Underground space and engineering feasibility are out of scope

12 AI scenario cards

S01	Industrial test & validation ground	Test
S02	Standards & compliance sandbox	Test
S03	Open-source collaboration station	Routine
S04	Robot delivery & low-speed shuttle	Test
S05	AI cultural guide	Routine
S06	Smart-device experience street	Routine
S07	Talent & enterprise service copilot	Routine
S08	AI+ health service navigation	Routine
S09	Accessibility & walkability diagnosis	Test
S10	Urban operations & public-safety review	Routine
S11	Visitor portal & multilingual reception	Routine
S12	Data-element & digital-asset services	Routine

Sensing scenarios must be human-reviewable, switchable off and explainable; none requires face recognition or identity profiling; a test scenario is not approved operation until a regulator says so.

One line, three cultural layers

Railway The engineering history and remains of the Jing-Zhang line — above all the switchback's habit of solving a hard problem with a clever structure, and the act of pinning a line to the ground.

Zhongguancun From an electronics street to an innovation ecosystem.

New AI culture Open-source collaboration, agent co-creation, memorable contribution.

The three are not parallel claims but one disposition in three eras: use a clever structure to turn an impossible gradient into a climbable path.

Events & operation

Ascent Week a city-scale open day; **Spike Day** the contributors' honour ceremony; **Switchback Talks** regular cross-disciplinary conversation.

Conversion path: visit a test ground → join a Switchback Talk → submit a contribution → receive a spike. Visitors move from watching to finding an interface they can act on.

All conceptual — not a government commitment, fixed programme, investment policy or funding arrangement.

Metrics, area recalculation & evidence chain

Every known metric is recomputable; every unknown one states what is missing

Fig. F5 · Metrics & evidence

METRICS · RECALCULATION · EVIDENCE CHAIN
18 known · 4 pending metrics · recalculated in EPSG:4548

Coverage

Announcement tasks 1.3–1.5 17 / 17

Agent tasks agent.1–agent.6 6 / 6

Mandatory standards 6 / 6

Core depth items 15 / 15

Four self-checks

Deterministic validation · trusted spatial review · visual packaging check · professional evidence review all PASS

Passing only means the package can enter machine checks and content review. It is not a judgement of quality and not an official approval.

Four metrics left unknown

Floor area ratio · height control · road density · residential capacity

Without approved regulatory conditions and baseline data, a precise-looking number is more dangerous than an empty one.

Evidence chain | from geometry to review

EVIDENCE CHAIN — GEOMETRY → METRICS → PROPOSAL → REVIEW

① Spatial evidence

geometry/*_geojson

site_boundary 1 · key_areas 3 (provisional)
land_use 187 (full cover, no overlap; key areas split to blocks)
buildings 320 · roads 19 · phasing 3
green_space 58 · public_space 19 · constraints 7

↓

② Recomputed metrics

metrics.json

Recomputed directly from the layers in EPSG:4548
18 known, each with formula, source files and confidence
4 unknown, each stating what is missing; nothing inferred

↓

③ Readable argument

proposal.md

[source:] [standard:] [depth:] [data:] [metric:]
Every required section carries machine-readable evidence tags
Five derived figures embedded in the report, counts consistent

↓

④ Response matrices

compliance / standard / depth

compliance_matrix 23 items (1.3–1.5 + agent.1–6)
standard_matrix 6 mandatory standards
design_depth_matrix 15 core depth items

↓

⑤ Self-check & review

self_check.json + repo scripts

deterministic validation · spatial review
visual packaging check · professional review
Human review and professional scoring rest with the reviewers

Nothing that cannot be recomputed from the data enters a conclusion.

Known metrics (18)

KNOWN METRICS — all recomputable from the submitted data

Site area

1 141.28 ha

site_area_sqm
polygon_area(site_boundary)
● confidence medium

Public-space ratio

7.23 %

public_space_ratio
union(PUBLIC_SPACE) / site_area
● confidence medium

Land-use parcels

187

land_use_parcel_count
count(LAND_USE)
● confidence high

Key-area footprints

308

key_area_building_count
count(BLDG in KEY_AREA)
● confidence high

Key areas

3

key_area_count
count(KEY_AREA)
● confidence high

Research area (announced)

43.6 km²

coordinated_research_area_sqm
Announced textual area
● confidence medium

AI scenario cards

12

scenario_card_count
count(scenario_cards)
● confidence high

Persona types

6

personas_count
count(personas)
● confidence high

Global ecosystem cases

7

ecosystem_case_count
count(cases)
● confidence medium

Green ratio

16.49 %

green_ratio
union(GREEN_SPACE) / site_area
● confidence medium

Building footprint area

94.29 ha

building_footprint_area_sqm
union(BUILDING_FOOTPRINT)
● confidence low

Key-area blocks

90

key_area_block_count
count(distinct block)
● confidence high

Spine walkway length

9 728.5 m

park_spine_length_m
length(ROAD-001)
● confidence medium

Key areas total (announced)

368.4 ha

key_area_total_sqm
Sum of announced areas
● confidence medium

Phasing stages

3

phasing_stage_count
count(PHASE)
● confidence high

Industrial test scenarios

4

test_validation_scenario_count
count(test_scenarios)
● confidence high

AI pilgrimage landmarks

4

ai_landmark_count
count(landmark_id)
● confidence high

Renewal project items

18

renewal_project_count
count(project list)
● confidence medium

Pending metrics (4)

UNKNOWN — data missing; nothing inferred

Floor area ratio

floor_area_ratio
No approved FAR condition

Height control

building_height_control_m
No height or heritage control

Road density

road_network_density...
No official road network or redlines

Residential capacity

resident_population_capacity
No baseline population, households or housing stock

All four stay unknown until official data arrives.

Self-check & coverage

SELF-CHECK & COVERAGE

- Deterministic validation PASS
- Trusted spatial review PASS
- Visual packaging check PASS
- Professional evidence review PASS

Announcement tasks 1.3–1.5 17 / 17

Agent tasks agent.1–6 6 / 6

Mandatory standards 6 / 6

Core depth items 15 / 15

Boundary still provisional_rough

⚠ Provisional rough boundary · not an official redline · recalculate every layer and metric once official polygons are released

Passing self-check only means the package can enter machine checks and content review. It is not a judgement of quality and not an official approval.

Fig. F5 | evidence chain, 18 known metrics, 4 pending metrics, self-check and coverage.